Operation Management

BITS Pilani
Pilani Campus

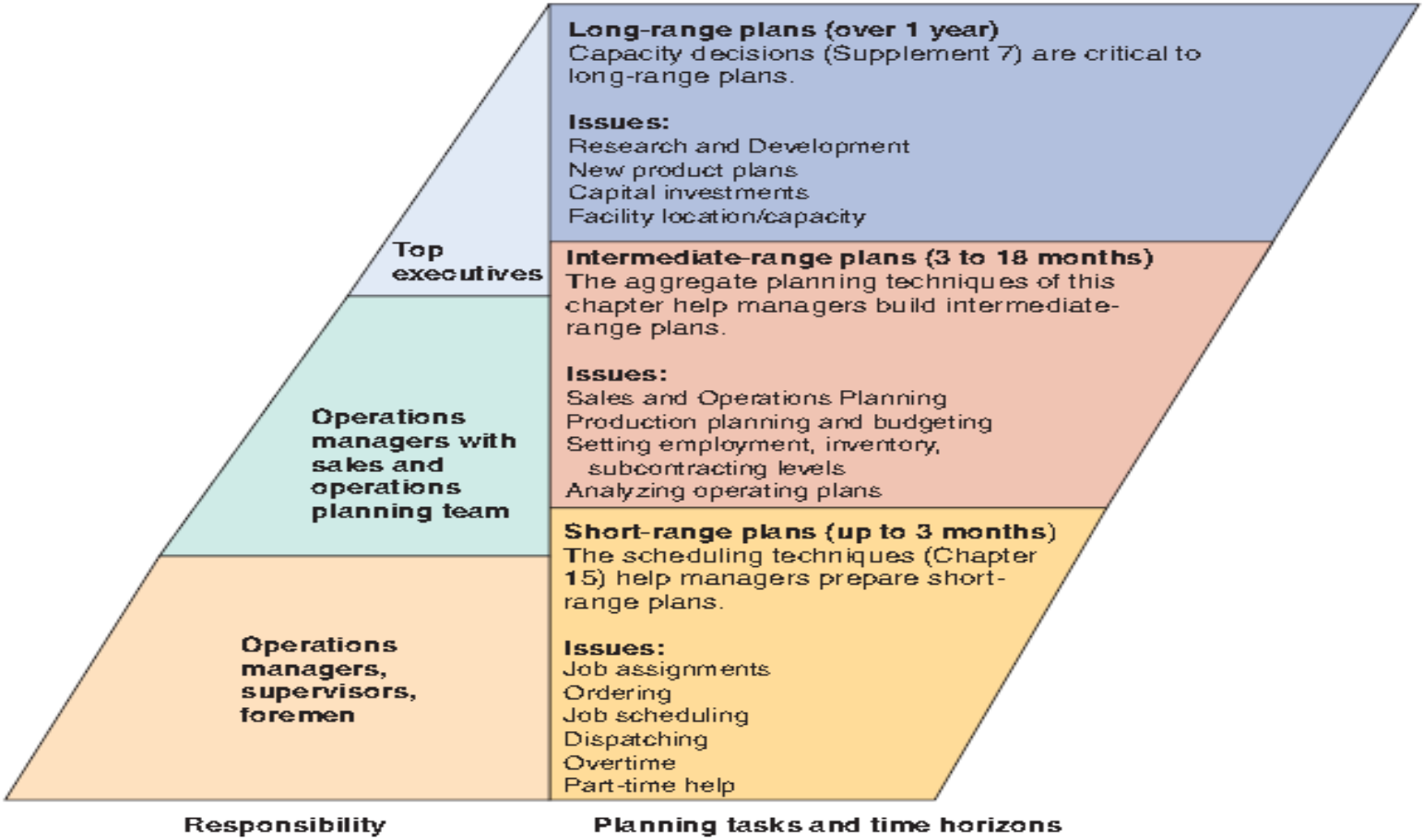
Abhishek Maiti

BITS Pilani
Pilani Campus

<MBAZG526/QMZG526, Operation Management>
Lecture No. 10

Planning Process

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Sales and Operation Planning

- *Intermediate planning*
- A process of balancing resources and forecasted demand.
- The output of S&OP is called an *aggregate plan*.
- Typically done by cross-functional teams.
- Determine which plans are feasible.
- Provides warning when resources do not match expectations.

Aggregate Planning

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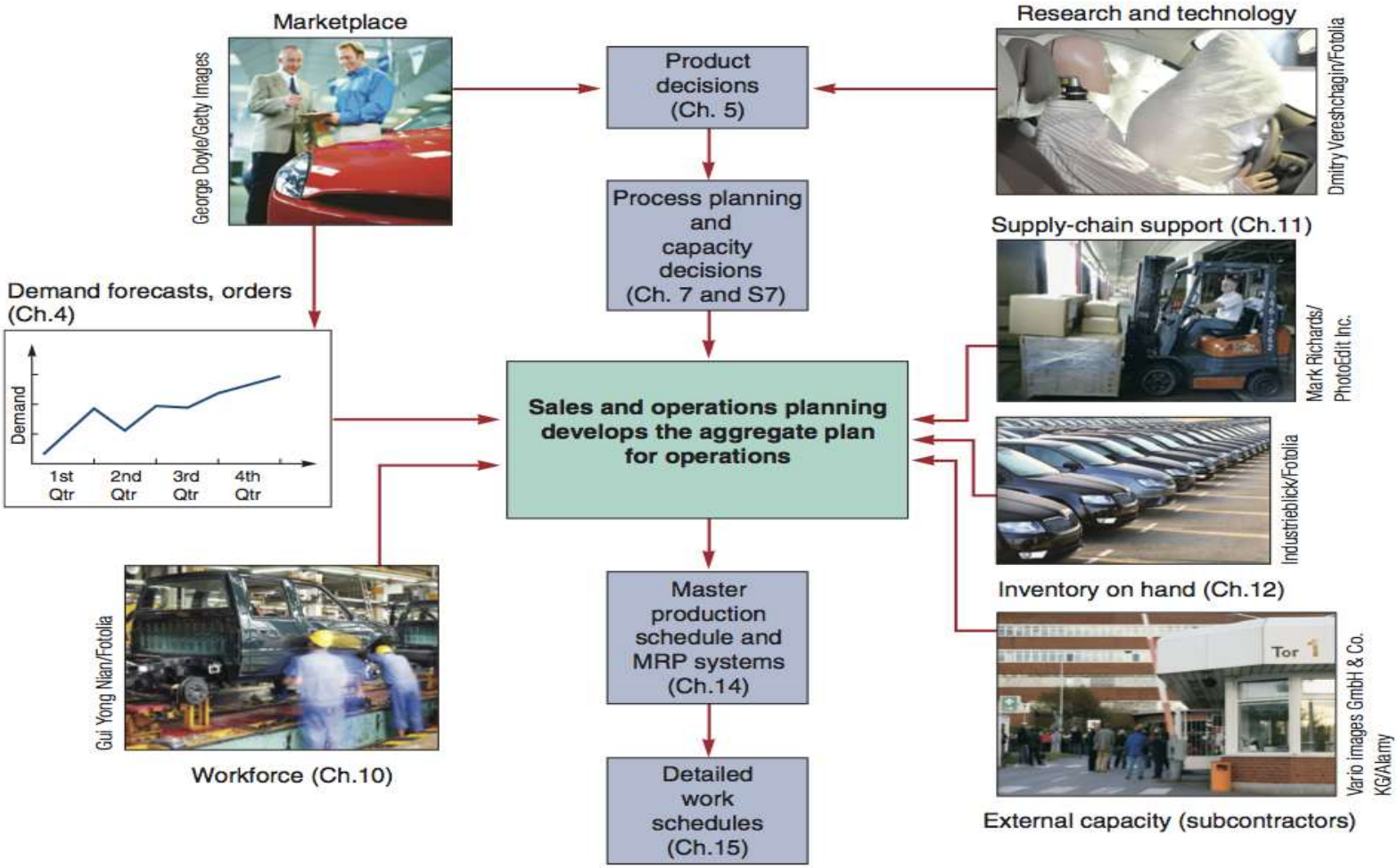

Nature of Aggregate Planning

- An S&OP team builds an aggregate plan that satisfies forecasted demand by adjusting production rates, labour levels, inventory levels, overtime work, subcontracting rates, and other controllable variables.
- *The objective of aggregate planning is usually to meet forecast demand while minimizing cost over the planning period.*

Nature of Aggregate Planning

- In a manufacturing environment, the process of breaking the aggregate plan down into greater detail is called disaggregation
- Disaggregation results in a master production schedule which provides input to material requirements planning (MRP) systems.

Aggregate Planning Strategies

- 8 Strategies (5 + 3)
- 5 Capacity Option Strategies and 3 Demand Option Strategies.

Capacity Options Strategies

- Changing inventory levels.
- Varying workforce size by hiring or layoffs
- Varying production rates through overtime or idle time
- Subcontracting
- Using part-time workers

Demand Options Strategies

- Influencing demand.
- Back ordering during high-demand periods.
- Counter-seasonal product and service mixing.

Advantages and Disadvantages

TABLE 13.1 Aggregate Planning Options: Advantages and Disadvantages

OPTION	ADVANTAGES	DISADVANTAGES	COMMENTS
Changing inventory levels	Changes in human resources are gradual or none; no abrupt production changes.	Inventory holding costs may increase. Shortages may result in lost sales.	Applies mainly to production, not service, operations.
Varying workforce size by hiring or layoffs	Avoids the costs of other alternatives.	Hiring, layoff, and training costs may be significant.	Used where size of labor pool is large.
Varying production rates through overtime or idle time	Matches seasonal fluctuations without hiring/training costs.	Overtime premiums; tired workers; may not meet demand.	Allows flexibility within the aggregate plan.
Subcontracting	Permits flexibility and smoothing of the firm's output.	Loss of quality control; reduced profits; potential loss of future business.	Applies mainly in production settings.
Using part-time workers	Is less costly and more flexible than full-time workers.	High turnover/training costs; quality suffers; scheduling difficult.	Good for unskilled jobs in areas with large temporary labor pools.
Influencing demand	Tries to use excess capacity. Discounts draw new customers.	Uncertainty in demand. Hard to match demand to supply exactly.	Creates marketing ideas. Overbooking used in some businesses.
Back ordering during high-demand periods	May avoid overtime. Keeps capacity constant.	Customer must be willing to wait, but goodwill is lost.	Many companies back order.
Counterseasonal product and service mixing	Fully utilizes resources; allows stable workforce.	May require skills or equipment outside firm's areas of expertise.	Risky finding products or services with opposite demand patterns.

Mixing Options to Develop a Plan

- A mixed strategy may be the best way to achieve minimum costs.
- There are many possible mixed strategies.
- Finding the optimal plan is not always possible.

Mixing Options to Develop a Plan

- **Chase strategy**
  - Match output rates to demand forecast for each period.
  - Vary workforce levels or vary production rate.
  - Favored by many service organizations.

Mixing Options to Develop a Plan

- **Level strategy**
  - Daily production is uniform
  - Use inventory or idle time as buffer
  - Stable production leads to better quality and productivity

Aggregate planning and S&OP

- **Hybrid Strategy:** A combination of chase demand and level production strategies. Offers a balance between adjusting production and maintaining a stable rate.
- **Backordering Strategy:** Allow demand to exceed supply temporarily, fulfilling backlogged orders later. Applicable when back-ordering is acceptable to customers.
- **Subcontracting Strategy:** Outsource production to third-party suppliers during periods of high demand. Useful for maintaining flexibility without expanding internal resources.

Source: Roger Schroeder | Susan Meyer Goldstein

Any Question

Methods for Aggregate Planning

- Graphical Methods.
- Transportation Problem.

Graphical Methods

- Graphical techniques are trial-and-error approaches that do not guarantee an optimal production plan, but they require only limited computations.
- A *cumulative* graph displays visually how the forecast deviates from the average requirements.

5 Steps of Graphical Methods

1. Determine the demand in each period.
2. Determine capacity for regular time, overtime, and subcontracting each period.
3. Find labour costs, hiring and layoff costs, and inventory holding costs.
4. Consider company policy that may apply to the workers or to stock levels.
5. Develop alternative plans and examine their total costs.

Class Exercise : Aggregate Planning for Roofing Supplier

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- A Juarez, Mexico, manufacturer of roofing supplies has developed monthly forecasts for a family of products.
- Data for the 6-month period January to June are presented in Table
- The firm would like to begin development of an aggregate plan.

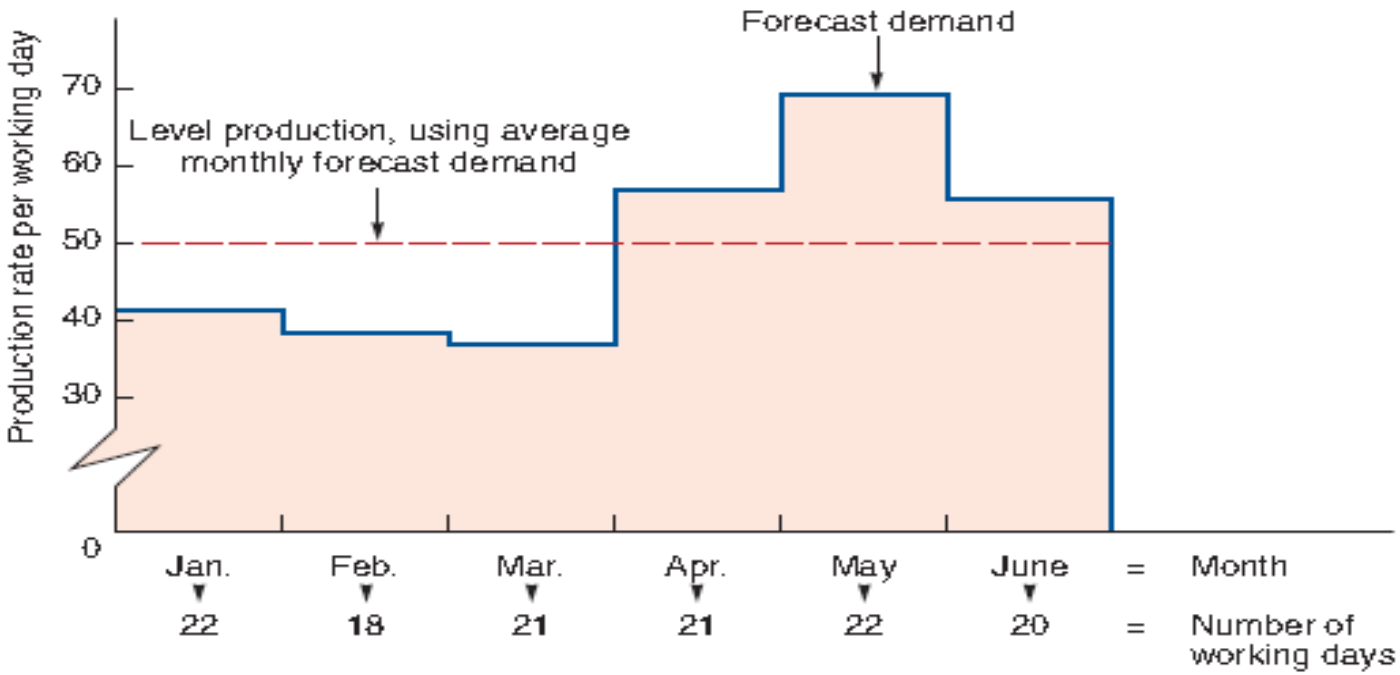
TABLE 13.2 Monthly Forecasts

MONTH	EXPECTED DEMAND	PRODUCTION DAYS	DEMAND PER DAY (COMPUTED)
Jan.	900	22	41
Feb.	700	18	39
Mar.	800	21	38
Apr.	1,200	21	57
May	1,500	22	68
June	1,100	20	55
	6,200	124	

Source: Heizer, Render, Muckson

Solution

$$\text{Average requirement} = \frac{\text{Total expected demand}}{\text{Number of production days}} = \frac{6{,}200}{124} = 50 \text{ units per day}$$



Source: Heizer, Render, Muckson

Plan

- One possible strategy (call it plan 1) for the manufacturer is constant workforce throughout the 6-month period.
- A second (plan 2) is to maintain a constant workforce at a level necessary to meet the lowest demand month (March) and to meet all demand above this level by subcontracting.
- Plan 3 is to hire and lay off workers as needed to produce exact monthly requirements.

TABLE 13.3 Cost Information

Inventory carrying cost	$ 5 per unit per month
Subcontracting cost per unit	$ 20 per unit
Average pay rate	$ 10 per hour ($80 per day)
Overtime pay rate	$ 17 per hour (above 8 hours per day)
Labor-hours to produce a unit	1.6 hours per unit
Cost of increasing daily production rate (hiring and training)	$300 per unit
Cost of decreasing daily production rate (layoffs)	$600 per unit

Source: Heizer, Render, Muckson

Plan 1

- Here we assume that 50 units are produced per day and that we have a constant workforce, no overtime or idle time, no safety stock, and no subcontractors.
- The firm accumulates inventory during the slack period of demand, January through March, and depletes it during the higher-demand warm season, April through June.

MONTH	PRODUCTION DAYS	PRODUCTION AT 50 UNITS PER DAY	DEMAND FORECAST	MONTHLY INVENTORY CHANGE	ENDING INVENTORY
Jan.	22	1,100	900	+200	200
Feb.	18	900	700	+200	400
Mar.	21	1,050	800	+250	650
Apr.	21	1,050	1,200	−150	500
May	22	1,100	1,500	−400	100
June	20	1,000	1,100	−100	0
					1,850

Total units of inventory carried over from one month to the next month = 1,850 units

Workforce required to produce 50 units per day = 10 workers

Source: Heizer, Render, Muckson

COST		CALCULATIONS
Inventory carrying	$ 9,250	(= 1,850 units carried × $5 per unit)
Regular-time labor	99,200	(= 10 workers × $80 per day × 124 days)
Other costs (overtime, hiring, layoffs, subcontracting)	0	
Total cost	$108,450	

Source: Heizer, Render, Muckson

Plan 2 : Use of Sub-contractor within a Constant Work Force

- Although a constant workforce is also maintained in plan 2, it is set low enough to meet demand only in March, the lowest demand-per-day month.
- To produce 38 units per day (800/21) in-house, 7.6 workers are needed.
- *All* other demand is met by subcontracting. Subcontracting is thus required in every other month. No inventory holding costs are incurred in plan 2.

SOLUTION ▶ Because 6,200 units are required during the aggregate plan period, we must compute how many can be made by the firm and how many must be subcontracted:

$$\text{In-house production} = 38 \text{ units per day} \times 124 \text{ production days}$$
$$= 4{,}712 \text{ units}$$
$$\text{Subcontract units} = 6{,}200 - 4{,}712 = 1{,}488 \text{ units}$$

The costs of plan 2 are computed as follows:

COST		CALCULATIONS
Regular-time labor	$ 75,392	(= 7.6 workers × $80 per day × 124 days)
Subcontracting	29,760	(= 1,488 units × $20 per unit)
Total cost	$105,152	

Source: Heizer, Render, Muckson

Plan 3 : Hiring and Layoff

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TABLE 13.4 Cost Computations for Plan 3

MONTH	FORECAST (UNITS)	DAILY PRODUCTION RATE	BASIC PRODUCTION COST (DEMAND ×1.6 HR PER UNIT × $10 PER HR)	EXTRA COST OF INCREASING PRODUCTION (HIRING COST)	EXTRA COST OF DECREASING PRODUCTION (LAYOFF COST)	TOTAL COST
Jan.	900	41	$14,400	—	—	$ 14,400
Feb.	700	39	11,200	—	$1,200 (= 2 × $600)	12,400
Mar.	800	38	12,800	—	$ 600 (= 1 × $600)	13,400
Apr.	1,200	57	19,200	$5,700 (= 19 × $300)	—	24,900
May	1,500	68	24,000	$3,300 (= 11 × $300)	—	27,300
June	1,100	55	17,600	—	$7,800 (= 13 × $600)	$ 25,400
			$99,200	$9,000	$9,600	$117,800

Source: Heizer, Render, Muckson

Plan 1 Vs 2 Vs 3

TABLE 13.5 Comparison of the Three Plans

COST	PLAN 1 (CONSTANT WORKFORCE OF 10 WORKERS)	PLAN 2 (WORKFORCE OF 7.6 WORKERS PLUS SUBCONTRACTORS)	PLAN 3 (HIRING AND LAYOFFS TO MEET DEMAND)
Inventory carrying	$ 9,250	$ 0	$ 0
Regular labor	99,200	75,392	99,200
Overtime labor	0	0	0
Hiring	0	0	9,000
Layoffs	0	0	9,600
Subcontracting	0	29,760	0
Total cost	$108,450	$105,152	$117,800

Source: Heizer, Render, Muckson

Mathematical Approaches

- **Useful for generating strategies**
  - Transportation Method of Linear Programming
    - Produces an optimal plan
    - Works well for inventories, overtime, subcontracting
    - Does not work when nonlinear or negative factors are introduced
- **Other Models**
  - General form of linear programming
  - Simulation

Class Exercise

- Farnsworth Tire Company would like to develop an aggregate plan via the transportation method.
- Data that relate to production, demand, capacity, and cost at its West Virginia plant

TABLE 13.6 **Farnsworth's Production, Demand, Capacity, and Cost Data**

	SALES PERIOD		
	MAR.	APR.	MAY
Demand	800	1,000	750
Capacity:			
Regular	700	700	700
Overtime	50	50	50
Subcontracting	150	150	130
Beginning inventory	100 tires		

COSTS	
Regular time	$40 per tire
Overtime	$50 per tire
Subcontract	$70 per tire
Carrying cost	$ 2 per tire per month

Source: Heizer, Render, Muckson

Solution

- See Excel

Minimize Cost

Period	Demand	Regular Capacity	Overtime Capacity	Subcontract Capacity
1	2000	1000	1000	5000
2	4500	1000	1000	5000
3	7500	1000	1000	5000
4	3000	1000	1000	5000

- Regular production cost per unit $20
- Overtime production cost per unit $25
- Subcontracting cost per unit $30
- Inventory holding cost per unit per period $ 3

Aggregate Planning in Services

Successful techniques for controlling the cost of labour in service firms include:

1. Accurate scheduling of labour-hours to ensure quick response to customer demand.
2. An on-call labour resource that can be added or deleted to meet unexpected demand.
3. Flexibility of individual worker skills that permits reallocation of available labour.
4. Flexibility in rate of output or hours of work to meet changing demand.

Five Service Scenarios

Restaurants

1) Smoothing the production process
2) Determining the optimal workforce size

Hospitals

– Responding to patient demand

National Chains of Small Service Firms

– Planning done at national level and at local level

Five Service Scenarios

Miscellaneous Services

- Plan human resource requirements
- Manage demand

Airline industry

- Extremely complex planning problem
  1) Number of flights
  2) Number of passengers on all flights
  3) Air and ground personnel at each hub and airport
  4) Allocation of seats to fare classes

Take Home Exercise : Example 6 (Text Book)

- Klasson and Avalon, a medium-size Tampa law firm of 32 legal professionals, wants to develop an aggregate plan for the next quarter.
- The firm has developed 3 forecasts of billable hours for the next quarter for each of 5 categories of legal business it performs.
- The 3 forecasts (best, likely, and worst) are shown.

TABLE 13.8 **Labor Allocation at Klasson and Avalon, Forecasts for Coming Quarter (1 lawyer = 500 hours of labor)**

FORECASTED LABOR-HOURS REQUIRED				CAPACITY CONSTRAINTS	
(1) CATEGORY OF LEGAL BUSINESS	(2) BEST (HOURS)	(3) LIKELY (HOURS)	(4) WORST (HOURS)	(5) MAXIMUM DEMAND FOR PERSONNEL	(6) NUMBER OF QUALIFIED PERSONNEL
Trial work	1,800	1,500	1,200	3.6	4
Legal research	4,500	4,000	3,500	9.0	32
Corporate law	8,000	7,000	6,500	16.0	15
Real estate law	1,700	1,500	1,300	3.4	6
Criminal law	3,500	3,000	2,500	7.0	12
Total hours	19,500	17,000	15,000		
Lawyers needed	39	34	30		

Revenue Management

- **Allocating resources to customers at prices that will maximize revenue**

1. Service or product can be sold in advance of consumption
2. Demand fluctuates
3. Relatively fixed resource (capacity)
4. Segmentable demand
5. Low variable costs high fixed costs

Revenue Management Approaches

- **Airlines, hotels, rental cars, etc.**
  - Tend to have predictable duration of service and use variable pricing to control availability and revenue
- **Movies, arenas, performing arts centers**
  - Tend to have predicable duration and fixed prices but use seating locations and times to manage revenue

Revenue Management Approaches

- **Restaurants, golf courses**
  - Generally have unpredictable duration of customer use and fixed prices, may use "off-peak" rates to shift demand and manage revenue

How Revenue Management Works

1. *Multiple pricing structures.*

2. *Forecasts of the use and duration of the use.*

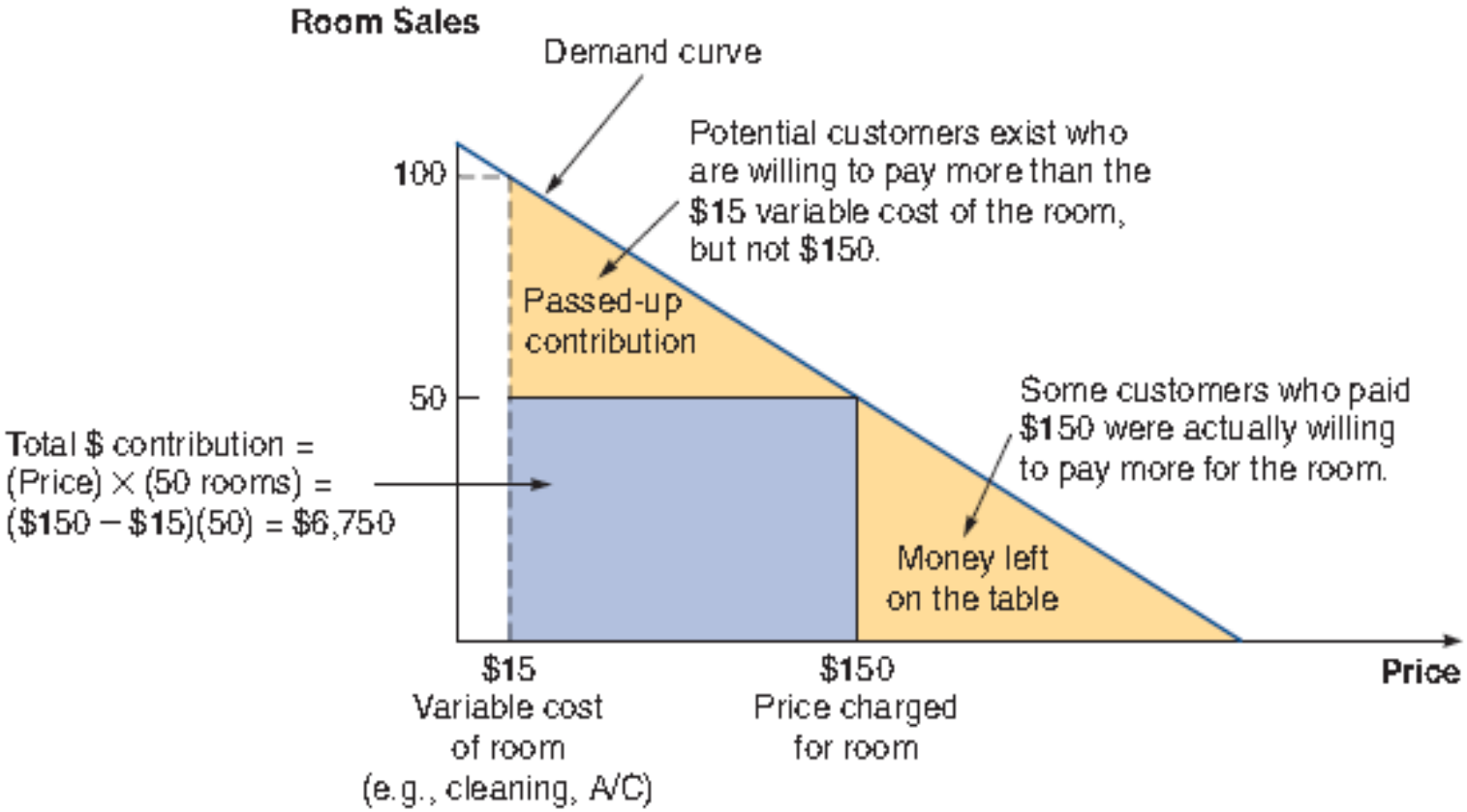
3. *Changes in demand.*

Class Exercise

- The Cleveland Downtown Inn is a 100-room hotel that has historically charged one set price for its rooms, $150 per night.
- The variable cost of a room being occupied is low.
- Management believes the cleaning, air-conditioning, and incidental costs of soap, shampoo, and so forth, are $15 per room per night.
- Sales average 50 rooms per night.
- Net sales are $6,750 per night with a single price point.

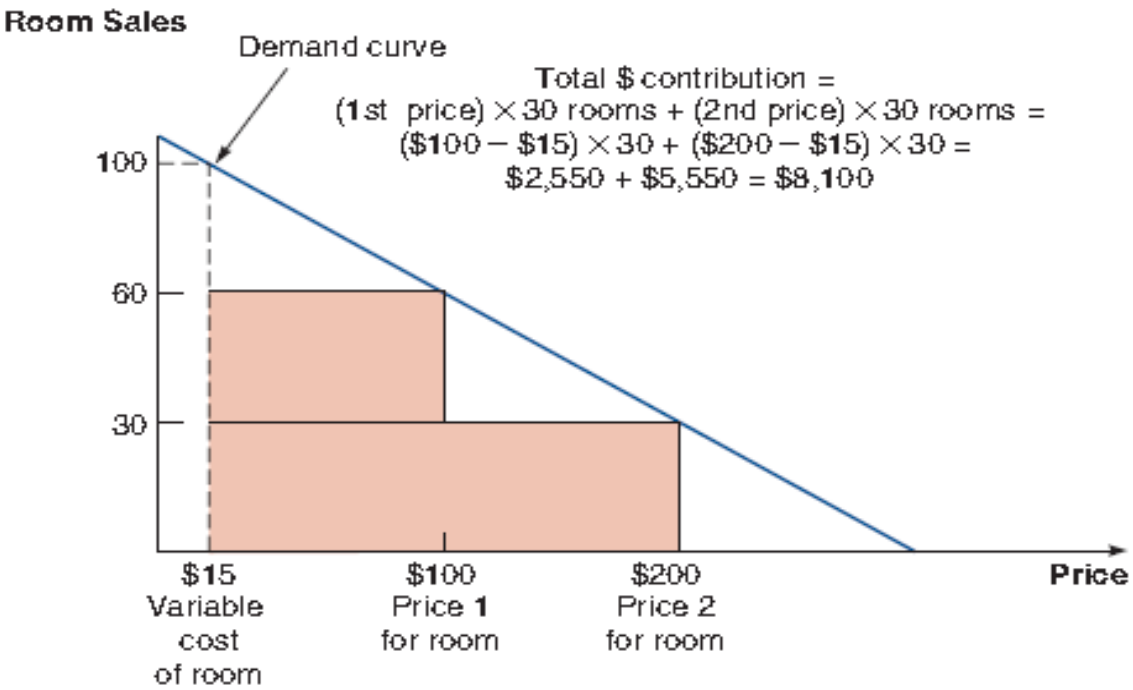
Solution

- Analyze pricing from the perspective of revenue management. We note that some guests would have been willing to spend more than $150 per room—“money left on the table.”
- Others would be willing to pay more than the variable cost of $15 but less than $150—“passed-up contribution.



Source: Heizer, Render, Muckson

- The inn decides to set *two* price levels. It estimates that 30 rooms per night can be sold at $100 and another 30 rooms at $200.



Source: Heizer, Render, Muckson

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 13

Any Question

THANK YOU